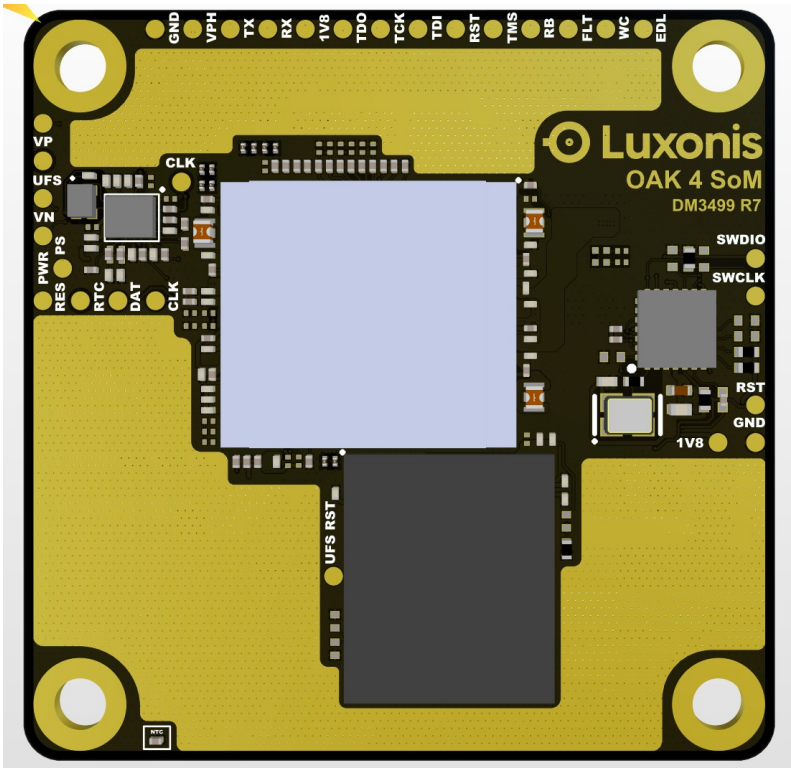
OAK4-SOM

System on Module

Technical Datasheet

Revision 2.2 | September 30, 2026



Datasheet compatibility
OAK-4-SOM

Preliminary information, subject to change without notice.
© 2026 Luxonis. All rights reserved.

Contents

Document Revision History 3
1 Product Overview 4
2 Key Specifications 5
3 Processing Hardware 6
4 Interfaces 7
5 Connector Pinout 8
6 Electrical Characteristics 15
7 Power and Thermal Characteristics 16
8 Mechanical Drawings 18
9 Package Contents 20
10 Documentation and Support 20

Document Revision History

Revision	Date	Author	Description of Changes
1.0	December 1, 2025	Luxonis	Initial release.
2.0	September 25, 2026	Luxonis	Updated styling; added hardware Factory Reset information and expanded electrical characteristics.
2.1	September 30, 2026	Luxonis	Aligned style and applicable processing, safety and support sections with OAK4-D and OAK4-S; reorganized interfaces and pinout.
2.2	September 30, 2026	Luxonis	Moved contents before document revision history.

1 Product Overview

OAK4-SOM combines a 6-core ARMv8 processor, RVC4 processing, 8 GB memory and 128 GB storage on a 40 × 40 mm module. Three mezzanine connectors link the module to a carrier board and expose camera, display, USB, PCIe and low-speed serial interfaces.

1.1 Product Configurations

SKU	Module configuration
OAK-4-SOM	RVC4; 8 GB LPDDR5x (PoP); 128 GB UFS

1.2 Module Layout

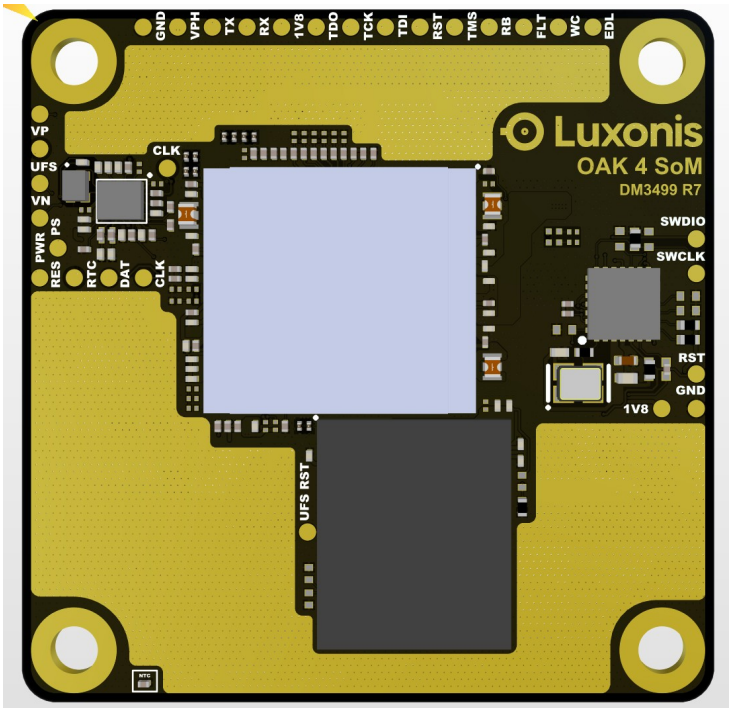


Figure 1: OAK4-SOM R7, component-side view.

2 Key Specifications

Parameter	Specification
Processor	Qualcomm QCS8550, 6-core ARM CPU
Memory	8 GB LPDDR5x (PoP)
Storage	128 GB UFS
AI Inference	Up to 52 TOPS (48 TOPS INT8 + 4 TOPS FP16)
Display interface	1 DSI, 4-lane D-PHY
Camera interfaces	5 CSI, 4-lane D-PHY; 2.5 Gbps per lane
Audio interface	1 I2S; 2 data lanes
USB	1 USB 3.1 Gen 1; up to 5 Gbps
PCIe Gen 4	1 interface, 2 lanes
PCIe Gen 3	1 interface, 2 lanes
SDIO	1 4-bit SDC (QSPI)
Dedicated UART	4 interfaces, including 1 debug UART
Dedicated I2C	4 interfaces
QUP	5 QUP (UART, I2C, SPI)
Dimensions	$40 \times 40 \times 2.9$ mm
Nominal module mass	5.7 g
Operating ambient temperature	-20 to $+50$°C
Power consumption	1.6 W (idle) to 25 W (peak)

3 Processing Hardware

3.1 RVC4 Processing Hardware

- Processor: Qualcomm QCS8550, 6-core ARMv8 CPU.
- System memory: 8 GB LPDDR5x (PoP).
- Onboard storage: 128 GB UFS.
- Operating system: Linux-based Luxonis OS, kernel 5.15.
- DSP arithmetic capacity: 48 TOPS INT8 or 12 TOPS FP16.
- GPU arithmetic capacity: 4 TOPS FP16.

The combined peak arithmetic capacity is 52 TOPS using DSP INT8 and GPU FP16 operations.

Computer Vision Engine

- Stereo depth processing: up to 800P at 60 FPS; 4-bit subpixel and 64-disparity search.
- Image warp: 1080P at 240 FPS for undistortion, resizing and cropping.
- Optical flow: semi-dense 1080P at 60 FPS or fully dense VGA at 60 FPS.
- Feature detection: Harris corners, 1080P at 60 FPS.
- Description matching: ORB calculation and inline matching with 256-bit descriptors.
- Template matching: up to 1080P; 1.2 ms for 500 templates.

Image Signal Processor (ISP)

- Up to three concurrent camera streams with ISP processing.
- Stream capacity: 3 × 36 MP at 30 FPS; 36 MP + 72 MP at 30 FPS; or 108 MP at 30 FPS.
- Up to two additional raw output streams without ISP processing.
- Automatic exposure, white balance and focus (3A).
- Hardware HDR: staggered HDR and digital overlap.
- Electronic image stabilization (EIS).
- Up to 18 bits per pixel.

Video Encoding and Decoding

- Encoding: 4K at 120 FPS, H.264 or H.265.
- Decoding: 4K at 240 FPS, H.264, H.265, VP9 or AV1.

4 Interfaces

4.1 Mezzanine Connectors

Parameter	Specification
Module connectors	3 Hirose mezzanine connectors
Module connector part number	DF40C-100DP-0.4V(51)
Carrier connector part number	DF40HC(3.0)-100DS-0.4V
Mated height	3 mm

Use the specified carrier connector for a 3 mm mated height. Alternative part numbers and mating heights have not been tested for functionality or fitment; the integrator is responsible for their use. Mechanical views appear in Section 8.

4.2 Interface Parameters

The interfaces are distributed across the three connectors to simplify carrier-board routing.

Interface	Specification
DSI	2 × 2-lane D-PHY or 1 × 4-lane D-PHY
CSI	CSI0, CSI2, CSI3, CSI4 and CSI5; 2.5 Gbps per lane
PCIe0	2 lanes, Gen 3; TX pairs are AC-coupled
PCIe1	2 lanes, Gen 4; TX pairs are AC-coupled
SDIO	SDC 3.0, up to 37.5 Mbps; 1.8 V logic
USB	USB 3.1 Gen 1; SuperSpeed, up to 5 Gbps
Debug UART	115200 baud; 8 data bits, no parity, 1 stop bit; no flow control

4.3 QUP Alternative Functions

QUP means Qualcomm Universal Peripheral. Each QUP serial engine (SE) supports one protocol at a time. L0–L6 identify the lines within an SE group.

Generic I/O	UART	I2C	SPI	I3C
QUP_L0	CTS	SDA	MISO	SDA
QUP_L1	RFR	SCL	MOSI	SCL
QUP_L2	Tx	-	SCLK	-
QUP_L3	Rx	-	CS_0	-
QUP_L4	-	-	CS_1	-
QUP_L5	-	-	CS_2	-
QUP_L6	-	-	CS_3	-

5 Connector Pinout

5.1 Connector Layout

Connector	Designator	Main interface groups
A	J1	Power supply and MIPI
B	J3	PCIe, QUP and peripheral interfaces
C	J4	MIPI, QUP and peripheral interfaces

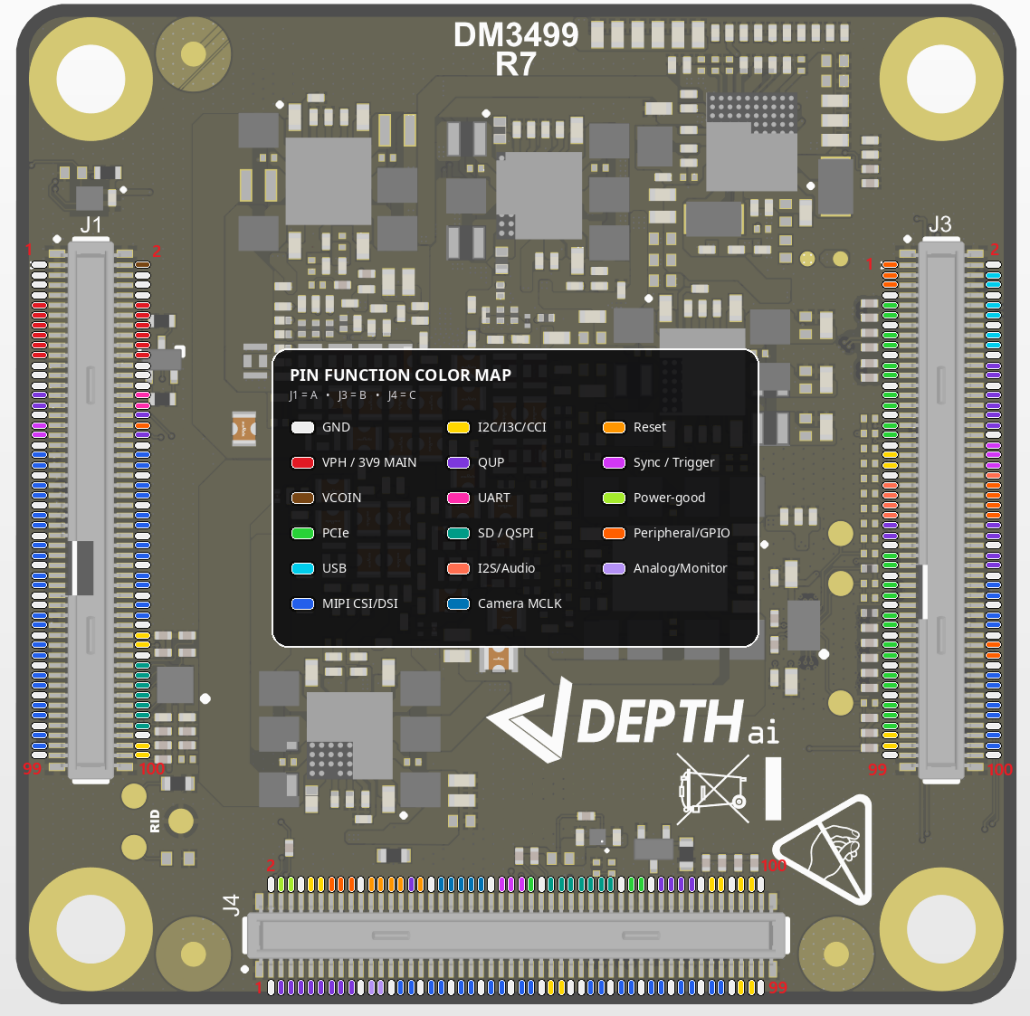


Figure 2: OAK4-SOM R7 pin-function groups, bottom view. Connector A is J1, B is J3 and C is J4.

Each connector has 100 signal contacts plus four corner mounting/soldering contacts. The illustration numbers identify the signal contacts.

5.2 Connector A (J1)

Pin No.	Pin Function	Description
1	GND	GND.
2	VCOIN_IN	Backup / coin-cell power input.
3	GND	GND.
4	GND	GND.
5	GND	GND.
6	GND	GND.
7	GND	GND.
8	GND	GND.
9	VPH	Main power, nominal 3.9 V.
10	VPH	Main power, nominal 3.9 V.
11	VPH	Main power, nominal 3.9 V.
12	VPH	Main power, nominal 3.9 V.
13	VPH	Main power, nominal 3.9 V.
14	VPH	Main power, nominal 3.9 V.
15	VPH	Main power, nominal 3.9 V.
16	VPH	Main power, nominal 3.9 V.
17	VPH	Main power, nominal 3.9 V.
18	VPH	Main power, nominal 3.9 V.
19	VPH	Main power, nominal 3.9 V.
20	VPH	Main power, nominal 3.9 V.
21	GND	GND.
22	GND	GND.
23	GND	GND.
24	GND	GND.
25	GND	GND.
26	GND	GND.
27	QUP1_SE0_L1	QUP1 SE0 L1.
28	DEBUG_UART_TX	Debug UART TX.
29	QUP1_SE0_L0	QUP1 SE0 L0.
30	DEBUG_UART_RX	Debug UART RX.
31	GND	GND.
32	QUP2_SE5_L2	QUP2 SE5 L2.
33	EXT_STRB	External STRB / trigger.
34	FORCE_USB_BOOT	Force USB boot.
35	EXT_FSYNC	External FSYNC.
36	QUP2_SE5_L3	QUP2 SE5 L3.
37	GND	GND.
38	GND	GND.
39	CSI5_LN0_P	MIPI CSI5 LN0+.
40	DSI0_LN0_P	MIPI DSI0 LN0+.
41	CSI5_LN0_N	MIPI CSI5 LN0-.
42	DSI0_LN0_N	MIPI DSI0 LN0-.
43	GND	GND.
44	GND	GND.
45	CSI5_LN1_P	MIPI CSI5 LN1+.
46	DSI0_LN1_P	MIPI DSI0 LN1+.
47	CSI5_LN1_N	MIPI CSI5 LN1-.
48	DSI0_LN1_N	MIPI DSI0 LN1-.
49	GND	GND.

Continued on next page

Pin No.	Pin Function	Description
50	GND	GND.
51	CSI5_CLK_P	MIPI CSI5 CLK+.
52	DSI0_CLK_P	MIPI DSI0 CLK+.
53	CSI5_CLK_N	MIPI CSI5 CLK-.
54	DSI0_CLK_N	MIPI DSI0 CLK-.
55	GND	GND.
56	GND	GND.
57	CSI5_LN2_P	MIPI CSI5 LN2+.
58	DSI0_LN2_P	MIPI DSI0 LN2+.
59	CSI5_LN2_N	MIPI CSI5 LN2-.
60	DSI0_LN2_N	MIPI DSI0 LN2-.
61	GND	GND.
62	GND	GND.
63	CSI5_LN3_P	MIPI CSI5 LN3+.
64	DSI0_LN3_P	MIPI DSI0 LN3+.
65	CSI5_LN3_N	MIPI CSI5 LN3-.
66	DSI0_LN3_N	MIPI DSI0 LN3-.
67	GND	GND.
68	GND	GND.
69	GND	GND.
70	DSI0_CLK1_P	MIPI DSI0 CLK1+.
71	CSI3_LN0_P	MIPI CSI3 LN0+.
72	DSI0_CLK1_N	MIPI DSI0 CLK1-.
73	CSI3_LN0_N	MIPI CSI3 LN0-.
74	GND	GND.
75	GND	GND.
76	SSC_I3C0_SCL	SSC I3C0 SCL.
77	CSI3_LN1_P	MIPI CSI3 LN1+.
78	SSC_I3C0_SDA	SSC I3C0 SDA.
79	CSI3_LN1_N	MIPI CSI3 LN1-.
80	GND	GND.
81	GND	GND.
82	SD_CARD_DET_N	SD card detect#.
83	CSI3_CLK_P	MIPI CSI3 CLK+.
84	SDC2_CLK_CONN	SDC2 CLK.
85	CSI3_CLK_N	MIPI CSI3 CLK-.
86	SDC2_CMD_CONN	SDC2 CMD.
87	GND	GND.
88	SDC2_DATA_0_CONN	SDC2 DATA0.
89	CSI3_LN2_P	MIPI CSI3 LN2+.
90	SDC2_DATA_1_CONN	SDC2 DATA1.
91	CSI3_LN2_N	MIPI CSI3 LN2-.
92	SDC2_DATA_2_CONN	SDC2 DATA2.
93	GND	GND.
94	SDC2_DATA_3_CONN	SDC2 DATA3.
95	CSI3_LN3_P	MIPI CSI3 LN3+.
96	GND	GND.
97	CSI3_LN3_N	MIPI CSI3 LN3-.
98	CCI_I2C5_SCL	CCI I2C5 SCL.
99	GND	GND.
100	CCI_I2C5_SDA	CCI I2C5 SDA.

5.3 Connector B (J3)

Pin No.	Pin Function	Description
1	LED_RGB_RED	RGB LED red.
2	GND	GND.
3	LED_RGB_GREEN	RGB LED green.
4	USB0_SS_TX0_P	USB0 SS TX0+.
5	LED_RGB_BLUE	RGB LED blue.
6	USB0_SS_TX0_N	USB0 SS TX0-.
7	GND	GND.
8	GND	GND.
9	PCIe1_QCS_RX0_P	PCIe1 QCS RX0+.
10	USB0_HS_D_P	USB0 HS D+.
11	PCIe1_QCS_RX0_N	PCIe1 QCS RX0-.
12	USB0_HS_D_N	USB0 HS D-.
13	GND	GND.
14	GND	GND.
15	PCIe1_QCS_RX1_P	PCIe1 QCS RX1+.
16	USB0_SS_RX0_P	USB0 SS RX0+.
17	PCIe1_QCS_RX1_N	PCIe1 QCS RX1-.
18	USB0_SS_RX0_N	USB0 SS RX0-.
19	GND	GND.
20	GND	GND.
21	PCIe1_QCS_REFCLK_P	PCIe1 QCS REFCLK+.
22	QUP1_SE2_L2	QUP1 SE2 L2.
23	PCIe1_QCS_REFCLK_N	PCIe1 QCS REFCLK-.
24	QUP1_SE2_L3	QUP1 SE2 L3.
25	GND	GND.
26	GND	GND.
27	PCIe1_QCS_TX0_N	PCIe1 QCS TX0-.
28	QUP1_SE1_L0	QUP1 SE1 L0.
29	PCIe1_QCS_TX0_P	PCIe1 QCS TX0+.
30	QUP1_SE1_L1	QUP1 SE1 L1.
31	GND	GND.
32	QUP1_SE1_L2	QUP1 SE1 L2.
33	PCIe1_QCS_TX1_N	PCIe1 QCS TX1-.
34	QUP1_SE1_L3	QUP1 SE1 L3.
35	PCIe1_QCS_TX1_P	PCIe1 QCS TX1+.
36	GND	GND.
37	GND	GND.
38	CSI4_STRB	CSI4 STRB.
39	I2C1_SCL/IO53	I2C1 SCL/IO53.
40	CSI5_FS	CSI5 FS.
41	I2C1_SDA/IO52	I2C1 SDA/IO52.
42	CSI0_FS	CSI0 FS.
43	GND	GND.
44	HIFI_DAC_I2S_MCLK	HiFi DAC I2S MCLK.
45	HIFI_DAC_I2S_WS	HiFi DAC I2S WS.
46	PoE_T2P	PoE T2P.
47	HIFI_DAC_I2S_DATA1	HiFi DAC I2S DATA1.
48	IR_LED_FAULT	IR LED FAULT.
49	HIFI_DAC_I2S_DATA2	HiFi DAC I2S DATA2.

Continued on next page

Pin No.	Pin Function	Description
50	IR_EN/PWM	IR EN/PWM.
51	HIFI_DAC_I2S_SCK	HiFi DAC I2S SCK.
52	IR_ADIM/HD	IR ADIM/HD.
53	QUP1_SE4_L2	QUP1 SE4 L2.
54	QUP1_SE4_L0	QUP1 SE4 L0.
55	QUP1_SE4_L3	QUP1 SE4 L3.
56	QUP1_SE4_L1	QUP1 SE4 L1.
57	GND	GND.
58	GND	GND.
59	PCIe0_QCS_TX0_N	PCIe0 QCS TX0-.
60	QUP1_SE5_L2	QUP1 SE5 L2.
61	PCIe0_QCS_TX0_P	PCIe0 QCS TX0+.
62	QUP1_SE5_L3	QUP1 SE5 L3.
63	GND	GND.
64	GND	GND.
65	PCIe0_QCS_TX1_N	PCIe0 QCS TX1-.
66	CSI0_LN2_P	MIPI CSI0 LN2+.
67	PCIe0_QCS_TX1_P	PCIe0 QCS TX1+.
68	CSI0_LN2_N	MIPI CSI0 LN2-.
69	GND	GND.
70	GND	GND.
71	PCIe0_QCS_REFCLK_N	PCIe0 QCS REFCLK-.
72	CSI0_LN3_P	MIPI CSI0 LN3+.
73	PCIe0_QCS_REFCLK_P	PCIe0 QCS REFCLK+.
74	CSI0_LN3_N	MIPI CSI0 LN3-.
75	GND	GND.
76	GND	GND.
77	PCIe0_QCS_RX0_N	PCIe0 QCS RX0-.
78	IMU1_LSM_INT/IO84	IMU1 INT/IO84.
79	PCIe0_QCS_RX0_P	PCIe0 QCS RX0+.
80	IMU2_LSM_INT/IO85	IMU2 INT/IO85.
81	GND	GND.
82	GND	GND.
83	PCIe0_QCS_RX1_N	PCIe0 QCS RX1-.
84	CSI0_CLK_P	MIPI CSI0 CLK+.
85	PCIe0_QCS_RX1_P	PCIe0 QCS RX1+.
86	CSI0_CLK_N	MIPI CSI0 CLK-.
87	GND	GND.
88	GND	GND.
89	PCIe0_WAKE_N	PCIe0 WAKE#.
90	CSI0_LN1_P	MIPI CSI0 LN1+.
91	PCIe0_CLK_REQ_N	PCIe0 CLKREQ#.
92	CSI0_LN1_N	MIPI CSI0 LN1-.
93	PCIe0_RESET_N	PCIe0 RESET#.
94	GND	GND.
95	CCI_I2C0_SCL	CCI I2C0 SCL.
96	CSI0_LN0_P	MIPI CSI0 LN0+.
97	CCI_I2C0_SDA	CCI I2C0 SDA.
98	CSI0_LN0_N	MIPI CSI0 LN0-.
99	GND	GND.
100	GND	GND.

5.4 Connector C (J4)

Pin No.	Pin Function	Description
1	GND	GND.
2	GND	GND.
3	QUP2_SE1_L0	QUP2 SE1 L0.
4	PG_3V3	3.3 V power-good.
5	QUP2_SE1_L1	QUP2 SE1 L1.
6	PG_1V8	1.8 V power-good.
7	QUP2_SE3_L2	QUP2 SE3 L2.
8	GND	GND.
9	QUP2_SE3_L3	QUP2 SE3 L3.
10	SSC_I2C1_SDA	SSC I2C1 SDA.
11	QUP2_SE7_L0	QUP2 SE7 L0.
12	SSC_I2C1_SCL	SSC I2C1 SCL.
13	QUP2_SE7_L1	QUP2 SE7 L1.
14	VCSEL_EN/PWM	VCSEL EN/PWM.
15	QUP2_SE7_L2	QUP2 SE7 L2.
16	VCSEL_ADIM/HD	VCSEL ADIM/HD.
17	QUP2_SE7_L3	QUP2 SE7 L3.
18	DOT_FAULT	Dot-projector fault.
19	GND	GND.
20	GND	GND.
21	PM_AMUX5	PM AMUX5.
22	CSI3_RESET_N	CSI3 RESET#.
23	PM_AMUX2	PM AMUX2.
24	CSI2_RESET_N	CSI2 RESET#.
25	GND	GND.
26	CSI4_AON_RESET_N	CSI4 AON RESET#.
27	CSI4_LN0_P	MIPI CSI4 LN0+.
28	CSI0_RESET_N	CSI0 RESET#.
29	CSI4_LN0_N	MIPI CSI4 LN0-.
30	QUP2_SE3_L0	QUP2 SE3 L0.
31	GND	GND.
32	CSI5_RESET_N	CSI5 RESET#.
33	CSI4_LN1_P	MIPI CSI4 LN1+.
34	GND	GND.
35	CSI4_LN1_N	MIPI CSI4 LN1-.
36	CSI0_MCLK	CSI0 MCLK.
37	GND	GND.
38	CSI4_AON_MCLK	CSI4 AON MCLK.
39	CSI4_CLK_P	MIPI CSI4 CLK+.
40	CSI1_MCLK	CSI1 MCLK.
41	CSI4_CLK_N	MIPI CSI4 CLK-.
42	CSI2_MCLK	CSI2 MCLK.
43	GND	GND.
44	CSI3_MCLK	CSI3 MCLK.
45	CSI4_LN2_P	MIPI CSI4 LN2+.
46	GND	GND.
47	CSI4_LN2_N	MIPI CSI4 LN2-.
48	CSI4_FS	CSI4 FS.
49	GND	GND.

Continued on next page

Pin No.	Pin Function	Description
50	CSI3_FS	CSI3 FS.
51	CSI4_LN3_P	MIPI CSI4 LN3+.
52	CSI2_FS	CSI2 FS.
53	CSI4_LN3_N	MIPI CSI4 LN3-.
54	PCIe1_WAKE_N	PCIe1 WAKE#.
55	GND	GND.
56	GND	GND.
57	CCI_AON_I2C_SCL	CCI AON I2C SCL.
58	SDC4_SIO0_QSPI_SIO0	SDC4 SIO0 / QSPI SIO0.
59	CCI_AON_I2C_SDA	CCI AON I2C SDA.
60	SDC4_SIO1_QSPI_SIO1	SDC4 SIO1 / QSPI SIO1.
61	GND	GND.
62	SDC4_SIO2_QSPI_SIO2	SDC4 SIO2 / QSPI SIO2.
63	GND	GND.
64	SDC4_SIO3_QSPI_SIO3	SDC4 SIO3 / QSPI SIO3.
65	CSI2_LN0_P	MIPI CSI2 LN0+.
66	QSPI_CS_0	QSPI CS 0.
67	CSI2_LN0_N	MIPI CSI2 LN0-.
68	SDC4_CLK_QSPI_CLK	SDC4 CLK / QSPI CLK.
69	GND	GND.
70	SDC4_CMD_QSPI_CS_1	SDC4 CMD / QSPI CS 1.
71	CSI2_LN1_P	MIPI CSI2 LN1+.
72	GND	GND.
73	CSI2_LN1_N	MIPI CSI2 LN1-.
74	PCIe1_RESET_N	PCIe1 RESET#.
75	GND	GND.
76	PCIe1_CLK_REQ_N	PCIe1 CLKREQ#.
77	CSI2_CLK_P	MIPI CSI2 CLK+.
78	GND	GND.
79	CSI2_CLK_N	MIPI CSI2 CLK-.
80	QUP2_SE6_L3	QUP2 SE6 L3.
81	GND	GND.
82	QUP2_SE6_L2	QUP2 SE6 L2.
83	CSI2_LN2_P	MIPI CSI2 LN2+.
84	QUP2_SE6_L0	QUP2 SE6 L0.
85	CSI2_LN2_N	MIPI CSI2 LN2-.
86	QUP2_SE6_L1	QUP2 SE6 L1.
87	GND	GND.
88	GND	GND.
89	CSI2_LN3_P	MIPI CSI2 LN3+.
90	I2C2_SDA	I2C2 SDA.
91	CSI2_LN3_N	MIPI CSI2 LN3-.
92	I2C2_SCL	I2C2 SCL.
93	GND	GND.
94	GND	GND.
95	CCI_I2C1_SDA	CCI I2C1 SDA.
96	CCI_I2C2_SCL	CCI I2C2 SCL.
97	CCI_I2C1_SCL	CCI I2C1 SCL.
98	CCI_I2C2_SDA	CCI I2C2 SDA.
99	GND	GND.
100	GND	GND.

6 Electrical Characteristics

6.1 Digital I/O Performance

Protocol-defined signals, including PCIe, MIPI and USB, follow their interface standards. Other GPIO signals use a 1.8 V logic level.

Parameter	Description	Min	Max	Unit
VIH	High-level input voltage, CMOS/Schmitt	0.7 x VDDPX_3	VDDPX_3 + 0.3	V
VIL	Low-level input voltage, CMOS/Schmitt	-0.3	0.3 x VDDPX_3	V
VSHYS	Schmitt hysteresis voltage	300	-	mV
VOH	High-level output voltage, CMOS	VDDPX_3 - 0.45	VDDPX_3	V
VOL	Low-level output voltage, CMOS	0.0	0.45	V
RPULL-UP	Pull-up resistance	20 K	60 K	Ohm
RPULL-DOWN	Pull-down resistance	20 K	60 K	Ohm

7 Power and Thermal Characteristics

7.1 Input Limits

VPH is the main module supply, nominally 3.9 V.

7.1.1 Absolute Maximum Ratings

Absolute limits are stress ratings. Exceeding them can permanently damage the module; operation at an absolute limit is not implied. Functionality and reliability are guaranteed only within the operating conditions in Section 7.1.2.

Parameter	Min	Nominal	Max	Unit
VPH	3.0	3.9	4.5	V

7.1.2 Operating Conditions

Parameter	Description	Min	Max	Unit
T_A	Ambient operating temperature	-20	50	°C
P_{PSE}	Power required on PSE side	12[a]	30	W

[a]Minimum power required to boot the device.

7.2 Power Consumption

Condition	Power	Notes
Idle (average)	1.6 W	Minimal processing
Boot (peak)	12 W	Initial power-on
Boot (average)	2.3 W	Typical 15 s boot sequence
Operating (average)	8–12 W	Application dependent
Operating (peak)	25 W	Maximum sustained load

Power consumption comes primarily from the Qualcomm QCS8550. AI workload power depends on model complexity and frame rate. Power dissipation also depends on cooling; insufficient cooling can reduce performance through thermal throttling at sustained peak loads.

7.3 Thermal and Electrical Safety

Subsystem integration

OAK4-SOM is a subsystem intended for integration into a complete product. It does not carry EMC, FCC or similar finished-product certification. The integrator is responsible for certifying the complete product.

Installation, mounting and connection must be performed by a skilled person. A skilled person has technical training and experience to recognize hazards and reduce risk. An ordinary person is neither skilled nor instructed.

ESD handling

The module is supplied in an ESD-safe bag. Handle it using appropriate ESD control procedures.

Hot surfaces

The top side of the module can reach 55°C or more during operation. Mount the module on a carrier board with an additional heat sink.

Ambient temperature	Expected maximum surface temperature
25°C (77°F)	55°C (131°F)
50°C (122°F)	80°C (176°F)

Caution: Hot surfaces can cause burns. Do not touch the module while it is powered. Disconnect power and allow the module to cool fully before handling.

8 Mechanical Drawings

8.1 Physical Dimensions

- PCB width: 40 mm
- PCB length: 40 mm
- Module thickness: 2.9 mm
- Nominal module mass: 5.7 g

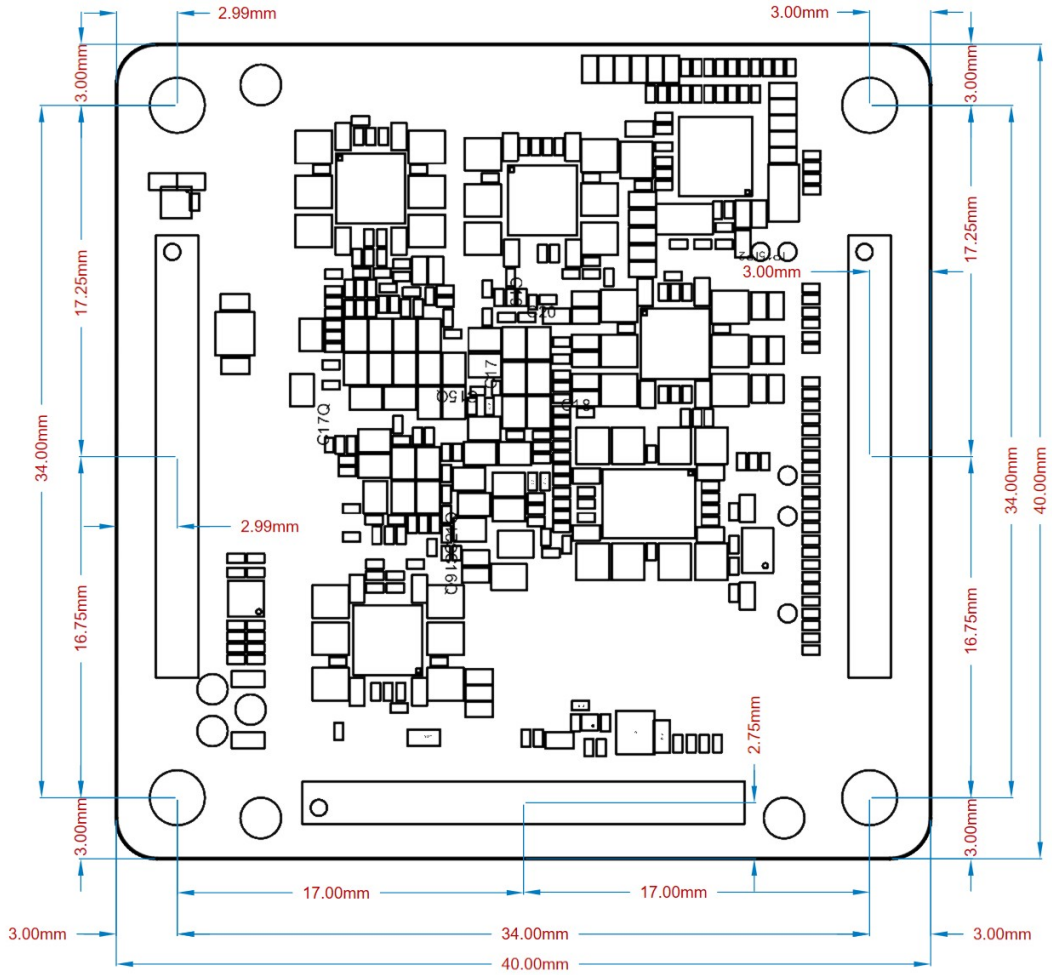


Figure 3: OAK4-SOM bottom view showing PCB outline, mounting and connector dimensions.

8.2 Module and Connector Locations

Use the component-side and side views with the bottom-view dimensions in Figure 3 when positioning the module, connectors and heat sink on the carrier board.

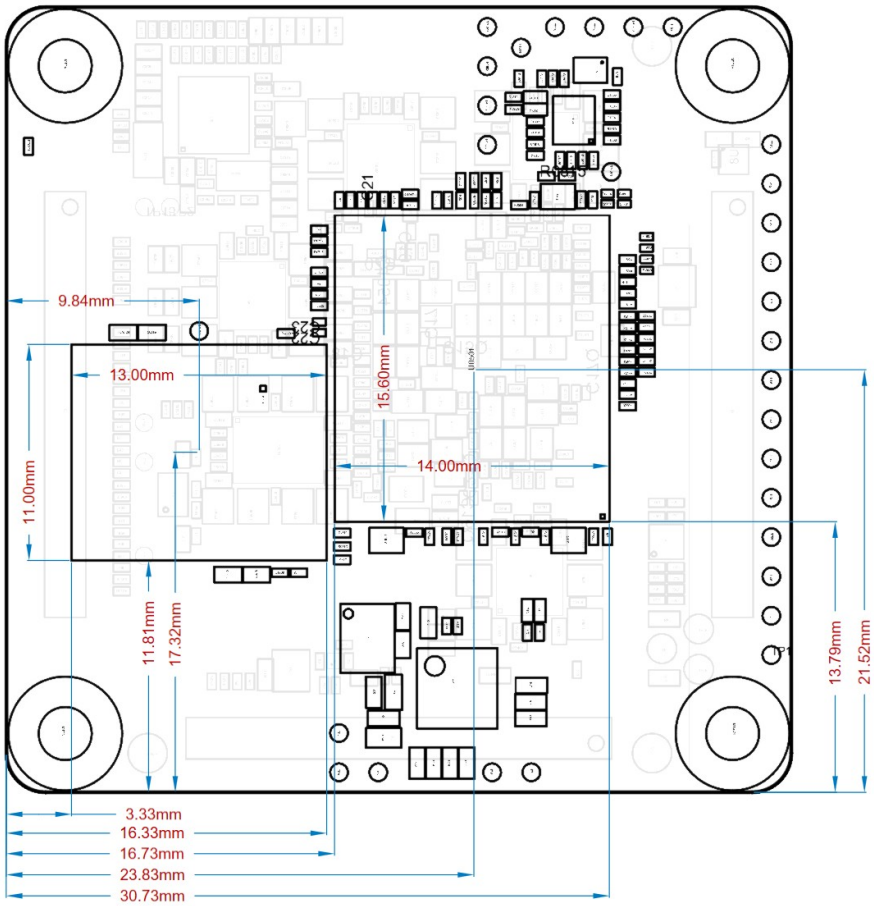


Figure 4: OAK4-SOM component-side dimensions.

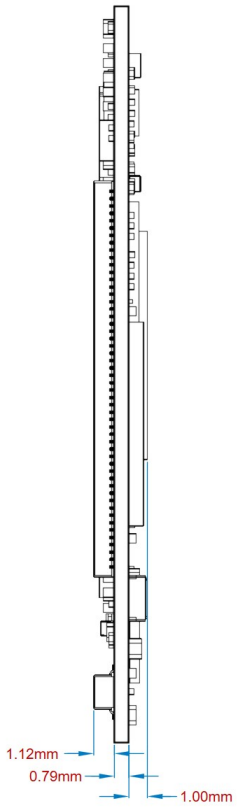


Figure 5: OAK4-SOM side view.

9 Package Contents

- OAK4-SOM module
- Get-started card

10 Documentation and Support

10.1 Online Resources

- OAK4-SOM Development Guide
- API Documentation

10.2 Technical Support

- Email: support@luxonis.com
- Forum: `https://discuss.luxonis.com`

Preliminary information, subject to change without notice.
© 2026 Luxonis. All rights reserved.